

Multiple Choice (10 marks)

1. Which of the following characteristics is predicted for an early-successional plant community?
 - (a) High detrital biomass
 - (b) High number of predatory species
 - (c) High presence of fully matured plant species
 - (d) High frequency of R-selected species
 - (e) High rates of soil nutrient depletion
2. Mitochondria isolated and placed in a buffered solution with a low pH begin to manufacture ATP. Which of the following is the best explanation for the effect of low external pH?
 - (a) It increases the acid concentration in the mitochondria matrix.
 - (b) It decreases the acid concentration in the mitochondria matrix.
 - (c) It increases the concentration of H^+ in the mitochondria matrix.
 - (d) It decreases the OH^- concentration in the mitochondria matrix.
 - (e) It increases diffusion of H^+ from the intermembrane space to the matrix.
3. In reference to a segment of DNA, which of the following molecules contains the fewest number of nucleotides?
 - (a) the tRNA transcript from the original DNA
 - (b) a single strand of the original DNA segment after a substitution mutation
 - (c) the final processed mRNA made from the original DNA
 - (d) the primary RNA transcript (after splicing) from the original DNA
 - (e) a single strand of the original DNA segment after a duplication mutation
4. All of the following might be found in connective tissue EXCEPT
 - (a) thrombin
 - (b) glycosaminoglycans
 - (c) collagens
 - (d) fibroblasts
5. What properties of water make it an essential component of living matter?
 - (a) Water is useful just because of its cooling properties.

- (b) Water is only important for hydration.
- (c) Water's role is limited to acting as a chemical reactant in cellular processes.
- (d) Water is important primarily because it contributes to the structural rigidity of cells.
- (e) Water's only function is to provide a medium for aquatic organisms.

True / False (10 marks)

Answer True or False

- 6. True or False: The neurilemma sheath fuses severed axon ends together to facilitate nerve regeneration.
- 7. True or False: The shuffling of alleles during sexual reproduction is the primary source of genetic variation in humans.
- 8. True or False: Removing the macronucleus from a paramecium would result in increased cell activity and growth.
- 9. True or False: The potato cells plasmolyze due to the influx of water from the surrounding environment.
- 10. True or False: A decrease in the amount of O₂ in the blood does not play a significant role in regulating respiration in humans.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. Discuss the implications of reduced hydrochloric acid secretion in the stomach on the digestive process, particularly regarding bacterial control and overall gut health.
- 12. Discuss the structure and function of the parathyroid glands in humans, emphasizing their role in calcium regulation and the implications of dysfunction in these glands.

End of Paper